Author Contributions Checklist Form

This form documents the artifacts associated with the article (i.e., the data and code supporting the computational findings) and describes how to reproduce the findings.

Part 1: Data

☐ This paper **does not** involve analysis of external data (i.e., no data are used or the only data are generated by the authors via simulation in their code).

☒ I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.

Abstract

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The analyses use the publicly available BlueBirds, CIFAR-10, and CIFAR-10H
datasets. The repository contains the label and derived-feature files used
directly in the analyses, links to the original data sources, and
documentation of the preprocessing steps.
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Availability

☒ Data **are** publicly available
☐ Data **cannot be made** publicly available

If the data are publicly available, see the *Publicly available data* section. Otherwise, see the *Non-publicly available data* section, below.

Publicly available data

☒ Data are available online at: https://github.com/Moriartycc/how-many-labelers-do-you-need
☐ Data are available as part of the paper's supplementary material.
☐ Data are publicly available by request, following the process described here:

☐ Data are or will be made available through some other mechanism, described here:

Non-publicly available data

Discussion of lack of publicly available data:

Description

File format(s)

☒ CSV or other plain text:

☒ Software-specific binary format (.Rda, Python pickle, etc.):

☐ Standardized binary format (e.g., netCDF, HDF5, etc.):

☐ Other (described here):

Data dictionary

☐ Provided by the authors in the following file(s):

☐ Data file(s) is (are) self-describiing (e.g., netCDF files)

☒ Available at the following URL:

https://github.com/Moriartycc/how-many-labelers-do-you-need

Additional information (optional)

The repository includes the exact label, feature, count, probability, and intermediate files used in the analyses. Experiment-specific README files describe each data file, its provenance, and its role in the workflow.

Part 2: Code

Abstract

The repository provides Python and MATLAB scripts for reproducing Figures 1, 2, and A.1. Experiment-specific README files identify the inputs, scripts, intermediate outputs, and final figures and provide step-by-step execution instructions.

Description

Code format(s)

☒ Script files
- ☐ R ☒ Python ☒ Matlab
- ☐ Other:

☐ Package
- ☐ R ☐ Python ☐ MATLAB toolbox
- ☐ Other:

☐ Reproducible report
- ☐ R Markdown ☐ Jupyter notebook
- ☐ Other:

☐ Shell script

☐ Other (described here):

Supporting software requirements

Version of primary software used

Python 3.11; MATLAB R2022b; CVX 2.2; MOSEK 9.3.22.

Libraries and dependencies used by the code

torch 2.6.0; torchvision 0.21.0; numpy 1.26.4; pandas 2.2.3; scipy 1.14.1; matplotlib 3.9.2; Crowd-Kit 1.4.2; cifar10_models at commit 641cac24371b17052b9bb6e56af1c83b5e97cd7f.

Supporting system/hardware requirements (optional)

A standard desktop is sufficient for plotting and smaller runs. A CUDA-capable GPU is strongly recommended for the full semisynthetic simulations. Allow several gigabytes of free disk space. MATLAB experiments run on a CPU with CVX and MOSEK. A valid MOSEK license is required.

Parallelization used

☒ No parallel code used
☐ Multi-core parallelization on a single machine/node
Number of cores used:
☐ Multi-machine/multi-node parallelization
Number of nodes and cores used:

License

☒ MIT License (default)
☐ BSD
☐ GPL v3.0
☐ Creative Commons
☐ Other (described here):

Additional information (optional)

Repository: https://github.com/Moriartycc/how-many-labelers-do-you-need. The repository includes an MIT LICENSE file.

Part 3: Reproducibility workflow

Scope

The provided workflow reproduces:
☐ Any numbers provided in text in the paper
☒ The computational method(s) presented in the paper (i.e., code is provided that implements the method(s))
☐ All tables and figures in the paper
☒ Selected tables and figures in the paper, as explained and justified here:

All computational figures (Figures 1, 2, and A.1) are covered. The paper contains no computational tables.

Workflow details

Format(s)

☐ Single master code file
☐ Wrapper (shell) script(s)
☐ Self-contained R Markdown file, Jupyter notebook, or other literate programming approach
☒ Text file (e.g., a readme-style file) that documents workflow
☐ Makefile
☐ Other (more detail in 'Instructions' below)

Instructions

The root README provides software setup and a figure-to-script map. Install the Python 3.11 dependencies with pip install -r requirements.txt, add the pinned PyTorch_CIFAR10 checkout and pretrained weights to PYTHONPATH, and configure MATLAB R2022b with CVX 2.2 and MOSEK 9.3.22. Run scripts from their experiment directories: code/bluebirds/main.m for Figure 1; the scripts in code/semisynthetic/ for Figure 2; and code/cifar10h/main.py followed by main.m for Figure A.1. Supplied CSV, MAT, and PDF outputs permit inspection without rerunning long simulations.

Expected run-time

Approximate time needed to reproduce the analyses on a standard desktop machine:
☐ <1 minute
☐ 1-10 minutes

☐ 10-60 minutes
☐ 1-8 hours
☒ >8 hours
☐ Not feasible to run on a desktop machine, as described here:

Additional documentation (optional)

The root README and the README file in each experiment directory document inputs, commands, outputs, runtime, hardware, and randomness.

Notes (optional)

Full simulation reruns are stochastic where noted. The repository archives the numerical outputs and final paper figures.